Chapter 10 Application to CEO Pay

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Statistical Decision Making (45-752)

1 Application to CEO Pay

1.0.1 Learning objectives

After completing this chapter, you will be able to:

- Analyze the relationship of CEO pay to firm market capitalization and number of employees.
- Use the US system for classifying industries.
- Develop a regression model that quantifies the relationship of CEO pay to firm market capitalization, number of employees, and industry.
- Apply visual diagnostics to assess the fit of a model to the data.
- Apply the Ramsey test to assess functional form.
- Critically evaluate regression models to identify strengths, weaknesses, and areas for improvement.

1.1 Introduction

This note set has broadly two objectives. One is to help you improve your skills in data analysis. The other is to help you learn about firms in the US economy. In particular, we will study the distribution of firm sizes as measured by two key variables, firm value (stock market capitalization) and number of employees. We will also study executive compensation, and we will investigate how executive compensation varies with firm value, number of firm employees, and the industry in which the firm operates. In this analysis, you will also learn about the US system for classifying industries. I think you will find this to be very interesting and informative.

[252]:
```
import tprstats
import pandas
import matplotlib.pyplot as plt
import numpy
from scipy import stats
import statsmodels.api as sm
import warnings
warnings.filterwarnings("ignore", category=RuntimeWarning)
```

```
# Jupyter "magic" command
# precision - round floats to 4 significant digits
%precision %.4f
```

[252]: '%.4f'

1.2 Overview of The Data

In preparing this data set, I assembled data for all US firms that have more than 99 employees and also have CEO compensation more than $99,000. The result is a total of 992 firms.

Sundar Pichai, the CEO of Google and Google's parent company Alphabet, receives an equity award every three years. He received nearly $226 million in 2022. He received just over $6 million in 2021 and $7.4 million in 2020. Hence, I averaged over the years 2020 through 2022, and recorded his 2022 pay as $79.8 million. Sources: https://www.cnn.com/2023/04/22/economy/alphabet-ceo-pay/index.html and https://observer.com/2023/04/google-ceo-sundar-pichai-2022-pay/

The CEO of Hertz, Stephen M. Scherr, will receive 182 million over five years if the stock performs well. Hence, I coded his annual salary as $36.4 million. The previous year, Hertz interim CEO Mark Fields was paid $14 million. I mention this to highlight that compensation of a company CEO can change very markedly from year to year, either because of a change in the CEO or for a change in compensation when there is not a change in CEO. The latter can occur in response to either unusually good or unusually poor firm performance. https://www.salary.com/tools/executive-compensation-calculator/hertz-global-holdings-inc-executive-salaries?year=2021. \

The CEO of Amazon, Andrew Jassy, received compensation of $1,298,723. In 2021, he received a stock award of $212 million vesting over 10 years. Hence, I divided the latter by 10 and added it to his 2022 compensation. Source: https://www.cbsnews.com/news/amazon-ceo-salary-andy-jassy-compensation/

The data are in a file named Mval_Comp_Emp_SIC.xlsx. Please import this data set. For convenience, I gave the file the shorthand name USFirms. You might want to do the same.

[253]:
```
# Import data set
USFirms = pandas.read_excel("data/CEO_Comp_Dat.xlsx")
```

For convenience for the analysis below, let's extract CEO Total Direct Compensation (TDC1), number of employees, market capitalization, gender, and Standard Industrial Capitalization 2-Digit codes (SIC2). We can then use the variables without having to including the names of the file in which the variables reside. Let's name the first three variables Comp, Emp, and MktCap respectively. In our data, both Comp and MktCap are expressed in millions of dollars. The Standard Industrial Capitalization codes are explained in the next section.

[254]:
```
Comp = USFirms["TDC1"]
Emp = USFirms["Employees"]
MktCap = USFirms["MktCap"]
Gender = numpy.where(USFirms["GENDER"] == "MALE", 0, 1)
SIC2 = USFirms["SIC2"]
YRS_CEO = USFirms["YRS_CEO"]
```

```
Age = USFirms["AGE"]
```

From the summary statistics below, we see the following. The average compensation of the CEO's in our sample was 9.56 million dollars. 75% of the CEO's made 12.047 million or more. The highest paid CEO earned 99.420 million. The company with the largest number of employes (Walmart) had 2.6 million employees. Market capitalization ranged from

[255]:
```
# Create a data frame with the variables shown below.
df = pandas.DataFrame({
    "Comp": Comp,
    "EMP": Emp,
    "MktCap": MktCap,
    "Gender": Gender,
    "YRS_CEO": YRS_CEO,
    "Age": Age
})
# Calculate summary statistics and display the result transposed
df.describe().T
```

[255]:

	count	mean	std	min	25%	\
Comp	992.0	9.560043	8.334033	0.245860	4.549702	
EMP	992.0	25691.151210	99350.687472	105.000000	2000.000000	
MktCap	992.0	25712.137051	129063.759285	4.581872	1772.953664	
Gender	992.0	0.071573	0.257909	0.000000	0.000000	
YRS_CEO	992.0	8.253024	7.335240	1.000000	3.000000	
Age	992.0	58.216734	6.748951	35.000000	54.000000	

	50%	75%	max
Comp	7.507077	12.047255	9.942010e+01
EMP	6300.000000	18700.000000	2.300000e+06
MktCap	4692.806656	15978.637824	2.750000e+06
Gender	0.000000	0.000000	1.000000e+00
YRS_CEO	6.000000	10.000000	4.300000e+01
Age	58.000000	62.000000	9.000000e+01

Let's calculate the combined market capitalization of the firms in our sample. We find that the 992 firms in our sample had combined market capitalization of $25.506 trillion. By way of comparison, the value of all US stocks at the end of 2022 was $40.512. Hence, the combined value of the 992 firms in our data set comprised roughly 63% of total stock market capitalization in the US.

Note the formatting of the following two commands.
MktCap.sum() calculates and prints the sum without rounding.
MktCap.sum().round(2) rounds to two digits after the decimal point.
It is often convenient to round variables.

Source: https://siblisresearch.com/data/us-stock-market-value/

[256]:
```
MktCap.sum()
```

[256]: 25506439.9546

[257]:
```
MktCap.sum().round(2)
```

[257]: 25506439.9500

The firms in our sample had a combined total of 25,485,522 employees. The total number of employed workers in the us at the end of 2022 was 159,244,000. Hence, the firms in our sample employed 16% of the US workforce.

Source https://www.bls.gov/news.release/empsit.t01.htm

[258]:
```
Emp.sum()
```

[258]: np.int64(25485622)

The command below calculates combined compensation of the 992 CEOs in our data. We obtain $9.484 billion.
For practice, modify the command to round to 3 digits after the decimal point.

[259]:
```
Comp.sum()
```

[259]: 9483.5627

[260]:
```
# Alternatively, you can use the following command to select the subset of␣
↪columns from the dataset
# and then use sum on the subset to get all sums at once.
USFirms[["TDC1", "Employees", "MktCap"]].sum()
```

[260]:
```
TDC1         9.483563e+03
Employees    2.548562e+07
MktCap       2.550644e+07
dtype: float64
```

The data set I assembled reports total compensation of each CEO. The data do not provide details of the form of CEO compensation. In most firms, a large proportion of CEO compensation comes in the form of vested stocks. Such vesting of stocks provides an incentive for the CEO to devote effort to increasing the market capitalization of the company, which in turn benefits the owners of the company, i.e., shareholders.

Source: If you wish to delve into the theoretical and empirical underpinnings of such compensation arrangements, I recommend: George-Levi Gayle, Chen Li, and Robert A. Miller, "How Well Does Agency Theory Explain Executive Compensation?," Federal Reserve Bank of St. Louis Review, Third Quarter 2018, 100(3), pp. 201-36. https://doi.org/10.20955/r.100.201-36

1.3 Standard Industrial Classification Codes

It is useful for you to have an understanding of the US Standard Industrial Classification (SIC) system. Each US firm is classified by the US government with a four-digit SIC code. The first two digits of the code are the industry in which the firm operates. The next two codes denote the particular focus of the firm. For our regression analysis below, we will use the two-digit codes.

The tables in Appendix A list the two-digit codes and the associated industries. The link in the following footnote leads you to the list of four-digit codes. I suggest you take a moment to view that list.

Source: https://www.bls.gov/oes/special-requests/oessic87.pdf

Here is an example. The primary four-digit code for Apple is 3663. The first two digits of this code, 36, and the table in Appendix A tell us that the industry in which Apple is classified is "Electronic and Other Electronic Equipment". If you go to the link in the footnote to the previous paragraph you will find that 3663 is: "Radio and Television Broadcasting and Communications Equipment". This classification is obviously due to Apple's focus on Communications Equipment. A second SIC code for Apple is is 3571, which is "Electronic Computers." The US government classification system only extends to four digits. You will sometime see codes with more than four digits. Such codes may be informative, but you should be aware that SIC codes with more than four digits are from a privately created coding system.

Appendix B provides a count of the number of firms in our sample that are in each SIC 2-digit industry. If you peruse this appendix, you will see that firms from 59 different industries are in our sample. You will also see that the industry with the largest number of firms is Business Services. It is entirely coincidental that the number of firms in Business Services, 73, is the same as SIC code for Business Services. You will also see that two firms have SIC code 99, which denotes Non-Classifiable Establishments. You might think these are obscure firms, but if you look at our data file, you will see that those two firms are Honeywell and General Electric.

Source: If you look at the web page for Honeywell, you will see the remarkably diverse range of products it sells: https://www.honeywell.com/us/en/industries/products. Likewise, if you go to General Electric's home page, you will see the remarkable range of businesses in which it operates: https://www.ge.com/

1.4 The Twelve Firms with Highest Market Capitalization

Let's investigate the data for the firms with the highest market capitalization. After some preliminary investigation with the command below, I found that there were 12 firms with market capitalization greater than 250 billion dollars. The first command below extracts the data for these 12 firms. The second command selects variables for the table presented below. We see that three firms had market capitalization greater than one trillion. In descending order of market capitalization, they are Apple, Microsoft, and Amazon. As you can see, the CEO's of these three companies received $99.4 million, $54.9 million, and $22.498 million. From the table, we also see that there is enormous variation across these twelve firms in number of employees, ranging from 2.3 million at Walmart to 20,000 at Broadcom.

By way of comparison, IBM had the highest market capitalization on January 1, 1991, $65.56 trillion. Adjusted for inflation to 2022 dollars, this would be $142.273 trillion.

[261]:
```
TopMktCap = USFirms.query("MktCap>250000")

print(TopMktCap[["CompanyName","MktCap","ExecName","TDC1","Employees","SIC2"]].
 ↪to_string(index=False))
```

```
CompanyName    MktCap            ExecName      TDC1  Employees  SIC2
      Apple 2750000.0        Timothy Cook 99.420097     164000    36
```

```
          Amazon 1210000.0        Andrew Jassy 22.498723    1541000    59
        Broadcom  283000.0         Hock E. Tan 60.606971      20000    36
      Home Depot  299000.0       Edward Decker 14.515600     500000    52
  JPMorgan Chase  408000.0         James Dimon 34.818729     293723    60
       Coca-Cola  272000.0       James Quincey 22.332484      79000    20
 Meta (Facebook)  633000.0     Mark Zuckerberg 27.110418      86482    73
       Microsoft 2370000.0       Satya Nadella 54.946309     221000    73
         Pepsico  264000.0 Ramon Luis Laguarta 24.136449     315000    20
Procter & Gamble  360000.0         Jon Moeller 17.716015     106000    28
            Visa  489000.0   Alfred Kelly, Jr. 28.103602      26500    60
         Walmart  409000.0    Douglas McMillon 24.642309    2300000    53
```

Together, the above 12 firms had a market capitalization of 9.75 trillion dollars. The total value of the US stock market was $40.5 trillion. Hence, the value of these 12 firms combined was 24% of the value of all US stocks!

[262]:
```
TopMktCap["MktCap"].sum()
```

[262]:
```
9747000.0000
```

1.5 Looking at Data

In your summer internship and your future work, you will very likely gather data (or have people who report to you gather data). When you have created a dataset, it is important for you to know how to "look at the data". We have already done so in previous applications, so the approach will be familiar to you. Also, we have already begun looking at the current data set in the investigations in the preceding sections of this chapter. We will now investigate the full sample in more detail. As you know, summary statistics, histograms, and graphs are useful ways to look at data.

1.6 Summary Statistics for The Full Sample

We see from the summary statistics below that the firm with the largest number of employees has 2.3 million employees. That firm is Walmart. We see that the firm with the largest market capitalization has value $2.75 trillion. That firm is Apple. The CEO of Apple received $99.42 million in 2022. While not shown in the summary statistics, the highest paid female executive was the CEO of Citigroup, Jane Nind Fraser, who earned $22.06 million in 2022. The median of CEO compensation is $7,507,078. Hence, half of the CEO's in our sample earned more than $7,507,078 and half earned less. We have an even number of observations. Hence, the reported median is between the pay of the two CEO's. One is the CEO of IDEX, a national securities exchange for trading of U.S. equities; the other is the CEO of TD SYNNEX, which provides information technology products and business process services. The lowest paid executive in this group is Bernard Francis Saul II who earned $245,860. He is the CEO of a real estate investment trust named Saul Centers Inc.

[263]:
```
USFirms[["TDC1", "Employees", "MktCap", "SIC2", "YRS_CEO", "AGE"]].describe().
 ↪round(3)
```

[263]:
```
           TDC1    Employees       MktCap     SIC2  YRS_CEO      AGE
count  992.000      992.000      992.000  992.000  992.000  992.000
```

```
mean      9.560    25691.151    25712.137   46.992    8.253   58.217
std       8.334    99350.687   129063.759   18.843    7.335    6.749
min       0.246      105.000        4.582   10.000    1.000   35.000
25%       4.550     2000.000     1772.954   35.000    3.000   54.000
50%       7.507     6300.000     4692.807   48.000    6.000   58.000
75%      12.047    18700.000    15978.638   62.000   10.000   62.000
max      99.420  2300000.000  2750000.000   99.000   43.000   90.000
```

For some of the analysis below, we will use logarithms of our three main variables. The logarithms are created here. We will also use the square of the logarithm of employment, so that variable is also calculated below.

[264]:
```
logComp = USFirms["TDC1"].apply(numpy.log)
logMktCap = USFirms["MktCap"].apply(numpy.log)
logEmp = USFirms["Employees"].apply(numpy.log)
logEmp_sq = logEmp.apply(numpy.square)
```

For illustrative purposes, here's an example of how you can take the log of multiple variables, relabeling the variables with the prefix "log_". This is not something you are expected to know for this course, but may be useful to you in the future when batch processing data. The first column in the output is Python's numbering of the rows in the data beginning with 0.

[265]:
```
mylogs = USFirms[["TDC1", "MktCap", "Employees"]].apply(numpy.log)
# prepend 'log_' to each column name
mylogs.columns = ["log_" + col for col in mylogs.columns]
# square of log of employment
mylogs["sq_log_Emp"] = mylogs["log_Employees"].apply(numpy.square)
mylogs.head()
```

[265]:

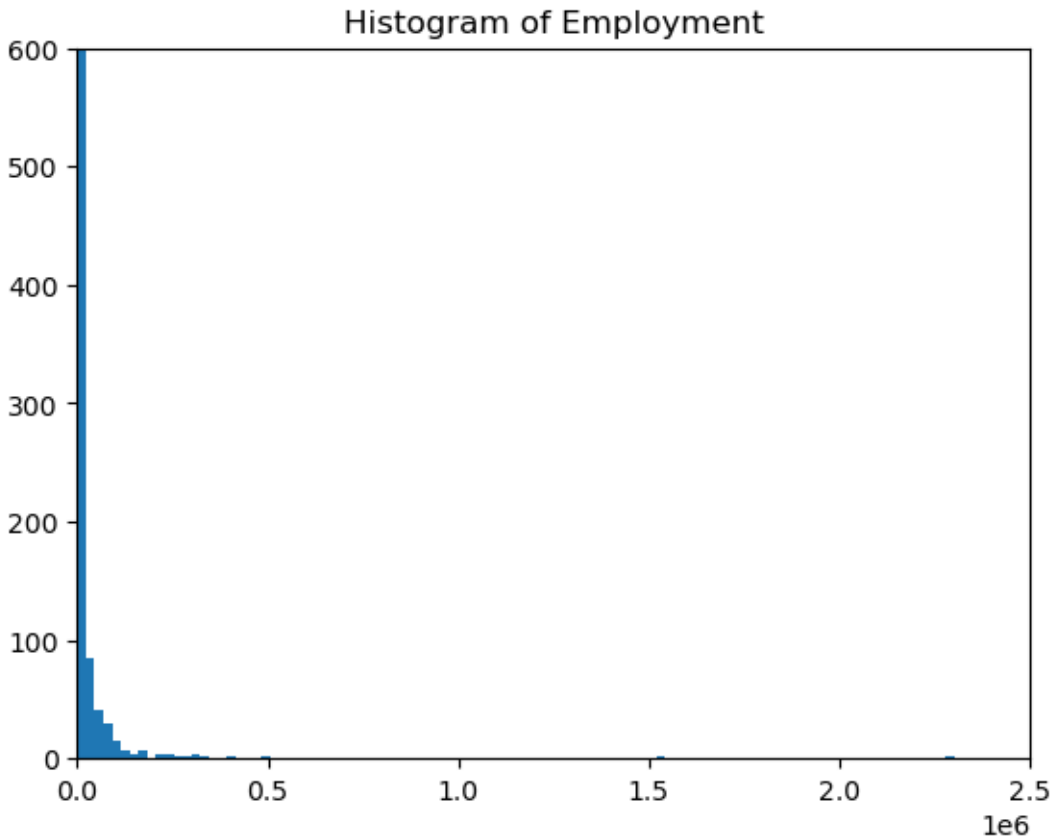
	log_TDC1	log_MktCap	log_Employees	sq_log_Emp
0	2.812255	10.550249	9.781320	95.674219
1	2.552363	8.789803	9.409191	88.532880
2	1.586507	9.192745	11.772979	138.603043
3	1.776470	5.995217	9.123693	83.241766
4	1.816332	8.509992	7.965893	63.455447

1.7 Investigation of the Distributions of The Variables

1.7.1 Distribution of Number of Employees

The distribution of number of employees is highly skewed to the right. This histogram is not very informative. Please do histograms of market capitalization and CEO compensation. You will see that those two variables are also highly skewed to the right.

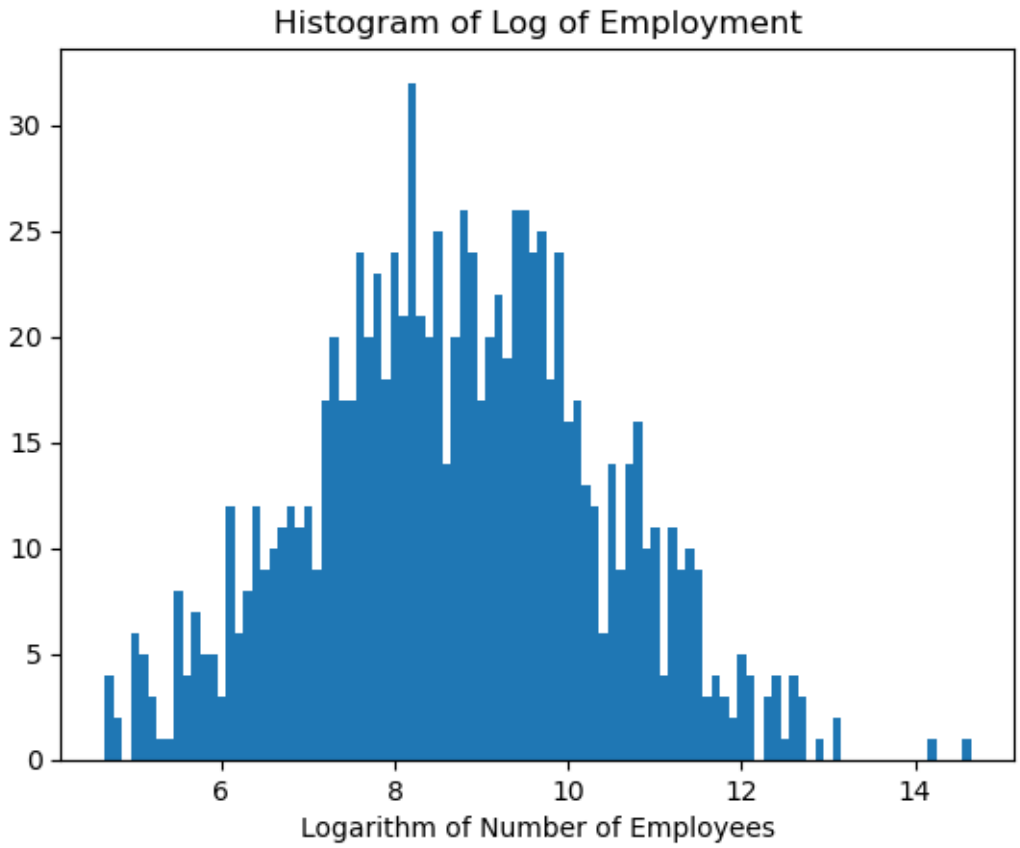
[266]:
```
plt.hist(Emp, bins=100)
plt.xlim(0, 2500000)
plt.ylim(0, 600)
plt.title("Histogram of Employment")
plt.show()
```



When working with skewed distributions, it is often informative to work with logs. We see that the histogram of the logarithm of employment is relatively symmetric.

Recall that logarithms can only be taken for numbers that are positive. All values are positive in our current application.

[267]:
```
plt.hist(logEmp, bins=100)
plt.xlabel("Logarithm of Number of Employees")
plt.title("Histogram of Log of Employment")
plt.show()
```

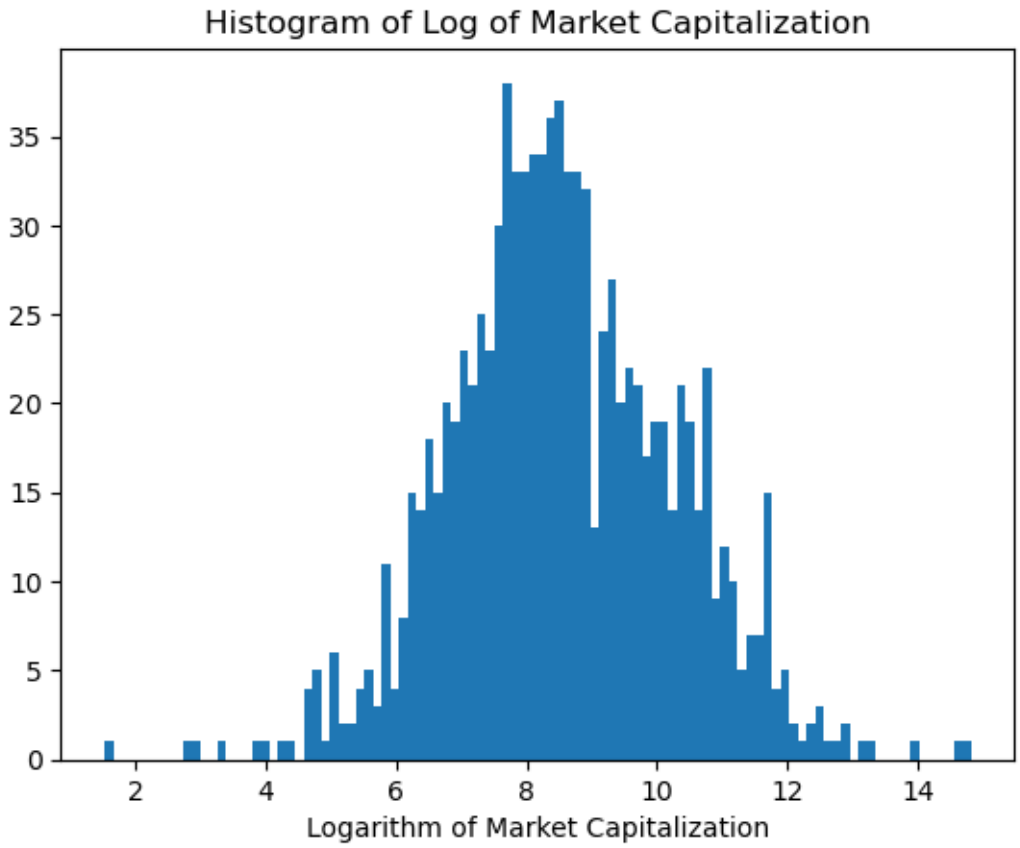


It is sometimes of interest to fit a distribution to a set of data. To introduce the approach, it is instructive to fit a distribution to the logarithms of each of our three variables. Let's begin by fitting a distribution to the data for the logarithm of number of employees. We can do so using the select_distribution() command. The select_distribution() command chooses the best fitting distribution from eight different distributions.

The normal distribution proves to be the best fitting for distribution for the logarithm of employment. When the log of a variable is normally distributed, we say that the variable itself has a lognormal distribution. Thus, either of the following is correct: In our sample of firms, the number of employees is lognormally distributed. In our sample of firms, the log of the number of employees is normally distributed.

1.7.2 Distribution of Market Capitalization.

[268]:
```
plt.hist(logMktCap, bins=100)
plt.xlabel("Logarithm of Market Capitalization")
plt.title("Histogram of Log of Market Capitalization")
plt.show()
```



In tprstats, we have incorporated a feature that fits a distribution to data. The next cell shows you how to use the command to fit a distribution to the logarithm of market capitalization. When you run the command, you will see that it chooses the t-distribution.

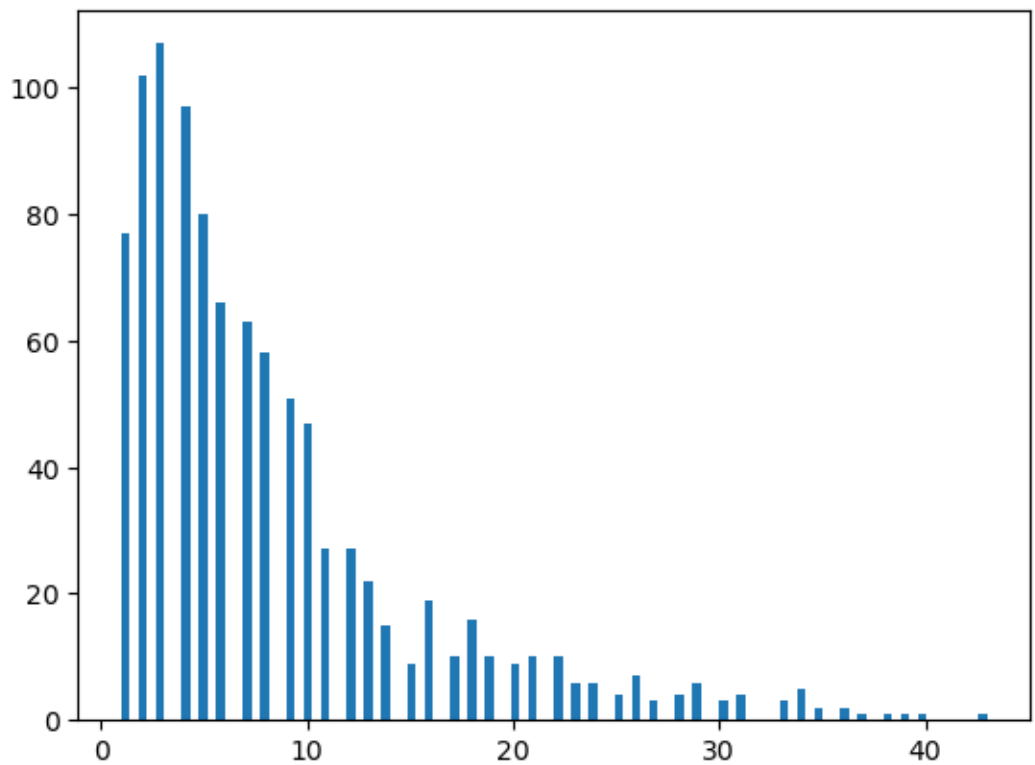
[269]:
```
tprstats.select_distribution(logMktCap)
```

[269]: 't'

We find that the logarithm of market capitalization has the t-distribution. Informally speaking, this tells us that there are relatively large outliers on the upper and lower ends of the distribution, more than we would obtain with a normal distribution. Either of the following is correct: In our sample of firms, MktCap has the log-t distribution; logMktCap has the t-distribution.

The variable YRS_CEO contains, for each of the 992 firms in our sample, the number of years that the firm's CEO has been CEO. The number of years as CEO is relatively skewed to the right.

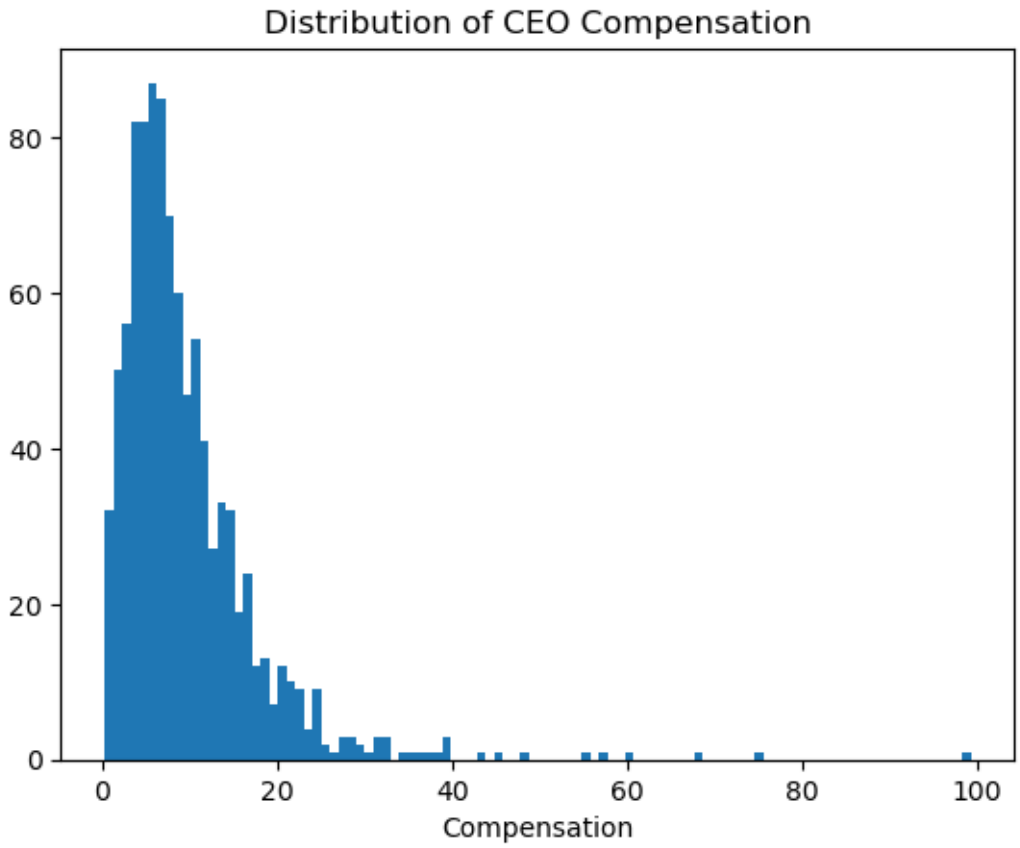
[270]:
```
plt.hist(YRS_CEO, bins=100)
plt.show()
```



1.7.3 Distribution of CEO Compensation

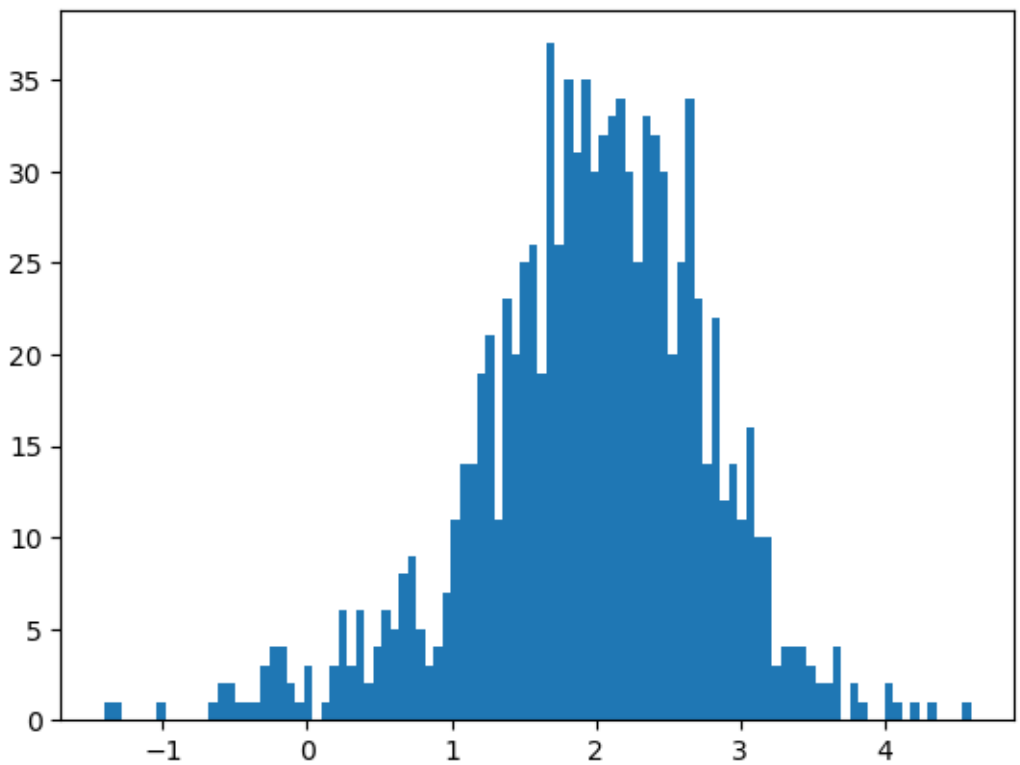
Not surprisingly, the distribution of CEO compensation is skewed to the right. We saw in our above summary table for the twelve firms with the highest market capitalization that some CEO's in our sample are very highly paid.

[271]:
```
plt.hist(Comp, bins=100)
plt.title("Distribution of CEO Compensation")
plt.xlabel("Compensation")
plt.show()
```



We see below that the distribution of the logarithm of CEO compensation is somewhat skewed to the left. Some CEOs in our sample have **relatively** low compensation. That said, we saw in the summary statistics that the lowest-paid CEO in our sample earned $245,860. This is high compensation relative to the vast majority of workers in the US.

[272]:
```
plt.hist(logComp, bins=100)
plt.show()
```



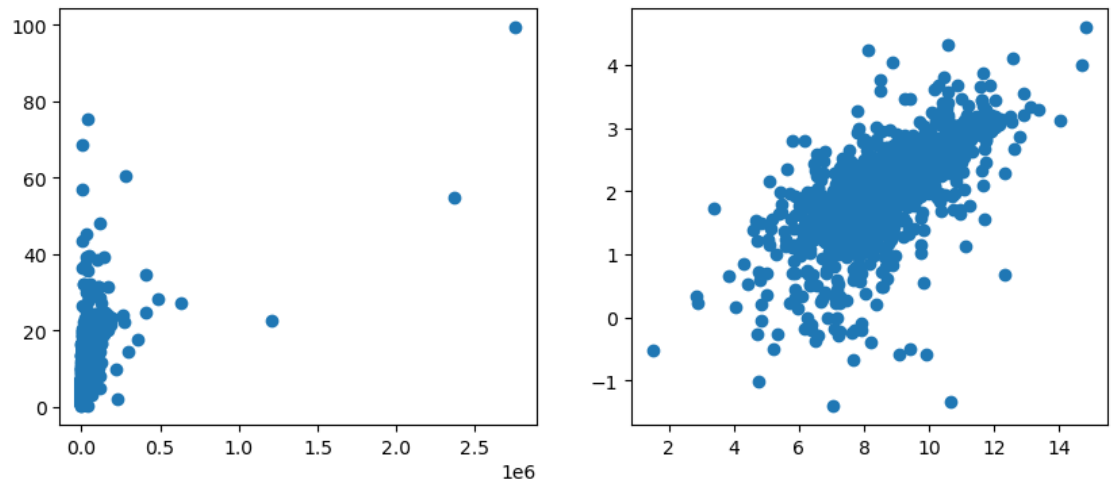
1.8 Analysis of Predictors of CEO Pay

1.8.1 Graphical Investigation

The objective of the analyses that follow is to investigate how much of the variation in CEO pay can be predicted using market capitalization and number of firm employees. We will begin by doing bivariate plots.

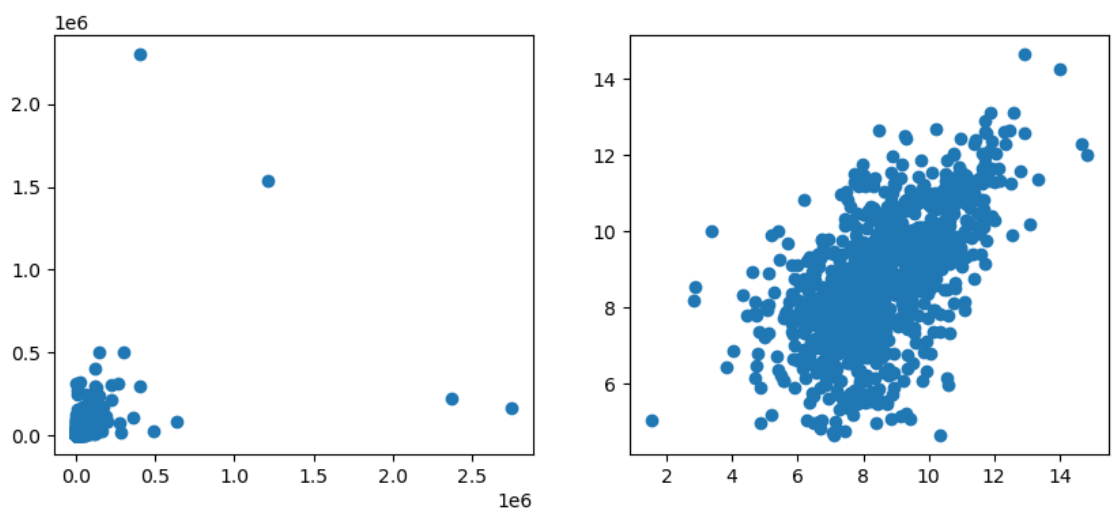
We first plot CEO compensation against market capitalization. The graph on the left below plots the unlogged variables. It is not very informative. The graph on the right plots the variables in logged form. In the latter we see that there is a strong tendency for CEO pay to increase with market capitalization. You are not required to know how to do the side-by-side plots, but I thought you might like to see how it is done. The subplots(1, 2, figsize=(10, 4)) command tells Python to place the plots in one row and two columns. The second and third commands create the plots.

[273]:
```
fig, axes = plt.subplots(1, 2, figsize=(10, 4))
axes[0].scatter(MktCap, Comp)
axes[1].scatter(logMktCap, logComp)
plt.show()
```



We next plot CEO compensation against number of employees. The graph on the left below plots the unlogged variables. It is not very informative. The graph on the right plots the variables in logged form. In the latter we see that there is a strong tendency for pay to increase with number of firm employees. The scatterplots above and below illustrate the usefulness of taking logarithms of variables.

[274]:
```
fig, axes = plt.subplots(1, 2, figsize=(10, 4))
axes[0].scatter(MktCap, Emp)
axes[1].scatter(logMktCap, logEmp)
plt.show()
```



1.9 Regression with Model Linear in Variables

The above bivariate graphs are of interest in giving us a feel for the data. To investigate the combined effect of market capitalization and firm employment in predicting CEO pay, we will use regression. Let's begin with a regression with CEO compensation as the dependent variable and firm employment and market capitalization as independent variables. From the p-values, we see that market capitalization is a significant predictor of CEO pay but number of employees is not.

[275]:
```
PayReg1 = tprstats.model("cs", "TDC1~Employees+MktCap", USFirms)
PayReg1.summary()
```

[275]:

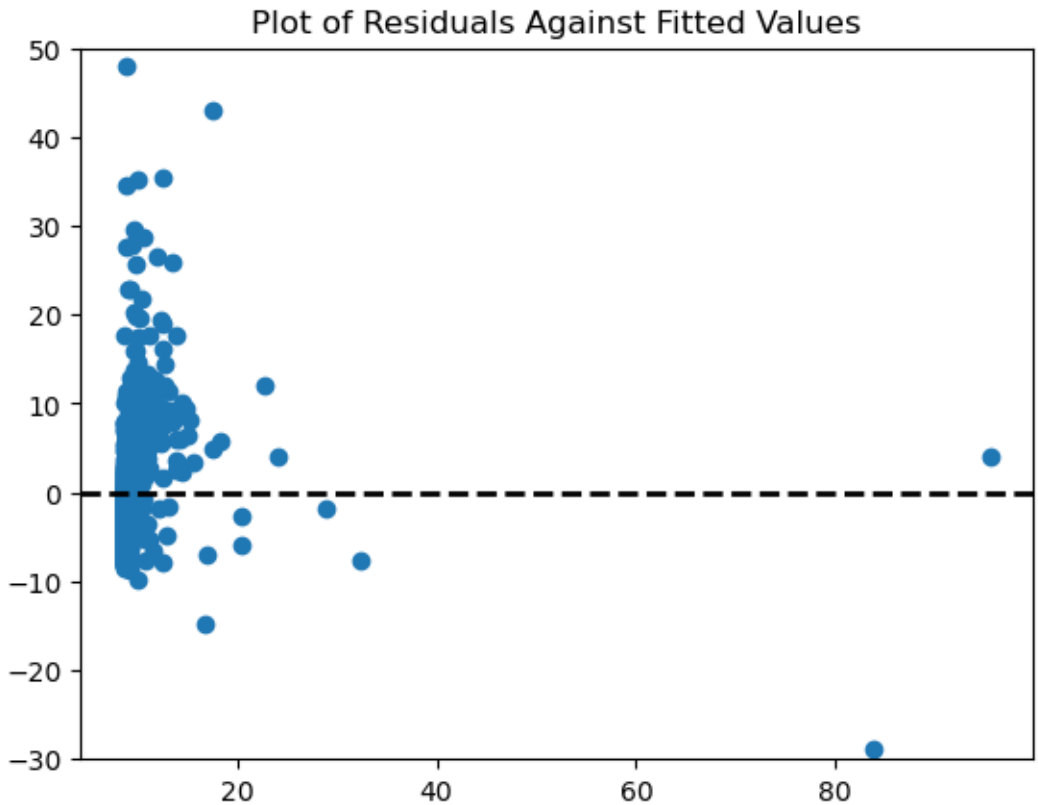
Dep. Variable:	TDC1	**R-squared:**	0.258
Model:	OLS	**Adj. R-squared:**	0.257
No. Observations:	992	**F-statistic:**	22.85
Covariance Type:	HC1	**Prob (F-statistic):**	1.99e-10

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	8.6338	0.251	34.428	0.000	8.142	9.126
Employees	4.74e-06	5.18e-06	0.915	0.360	-5.42e-06	1.49e-05
MktCap	3.129e-05	4.84e-06	6.461	0.000	2.18e-05	4.08e-05

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
[2] The condition number is large, 1.48e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

Is the above regression a satisfactory functional form? To investigate, let's extract and plot the residuals against the fitted values. The dashed line at zero shows the mean of the residuals. The pattern of the residuals strongly suggests that this model is not satisfactory.

[276]:
```
resid=PayReg1.resid
yhat=PayReg1.predict()
plt.scatter(yhat, resid)
plt.axhline(0, linestyle="--", linewidth=2, color="black")
plt.title("Plot of Residuals Against Fitted Values")
plt.ylim(-30, 50)
plt.show()
```



To investigate further, let's perform the Ramsey test. The extremely low p-values for Ramsey test reinforce the conclusion that this is not a satisfactory functional form.

[277]:
```
PayReg1.ramsey_test()
```

[277]:

	power	pvalue
0	2	1.310614e-20
1	3	8.953337e-49

1.10 Regression with Variables Expressed in Logarithmic Form

Let's now use logarithms of the variables. You know from our previous work how to interpret the coefficients of a model with variables expressed in log form; the coefficients are elasticities. Hence, we can interpret the coefficients below as follows. Holding market capitalization constant, a one percent increase in number of employees predicts a .1 percent increase in CEO pay. Holding number of employees constant, a one percent increase in market capitalization predicts a .25% increase in CEO pay. As shown by the R^2 statistic, this regression can account for 45% of the variation in CEO pay.

[278]:
```
PayReg2=tprstats.model("cs", "log(TDC1)~log(Employees)+log(MktCap)", USFirms)
PayReg2.summary()
```

[278]:

Dep. Variable:	log(TDC1)	**R-squared:**	0.450
Model:	OLS	**Adj. R-squared:**	0.449
No. Observations:	992	**F-statistic:**	372.0
Covariance Type:	HC1	**Prob (F-statistic):**	3.52e-121

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-1.0706	0.124	-8.645	0.000	-1.314	-0.828
log(Employees)	0.0997	0.016	6.129	0.000	0.068	0.132
log(MktCap)	0.2523	0.015	17.320	0.000	0.224	0.281

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Let's calculate and interpret the standardized coefficients. We see that, holding number of employees constant, a one standard deviation increase in the log of market capitalization predicts an increase of .527 standard deviations of the logarithm of compensation. Holding market capitalization constant, a one standard deviation increase in the log of number of employees predicts an increase of .204 standard deviations of the logarithm of compensation.

[279]:
```
PayReg2.standardized_coefficients()
```

[279]:
```
log(Employees)    0.204236
log(MktCap)       0.527684
dtype: float64
```

Thus, both the elasticities and the standardized coefficients imply that, within our sample, market capitalization of a firm has a substantially larger impact on CEO compensation than number of firm employees.

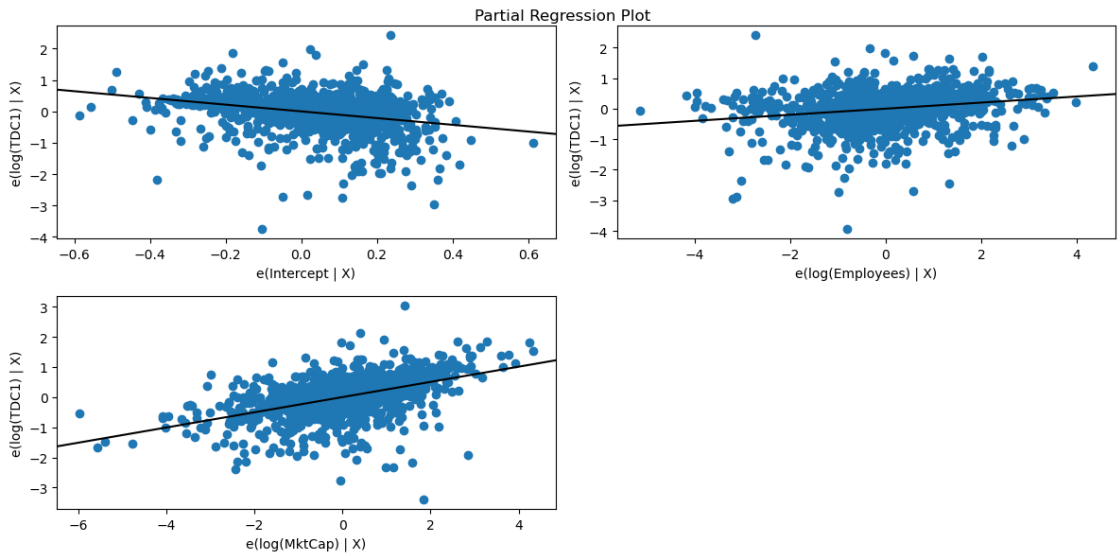
The p-values below for the Ramsey test are substantially higher for the above regression, but the p-value with one fitted term is still relatively low.

[280]:
```
PayReg2.ramsey_test()
```

[280]:

	power	pvalue
0	2	0.060853
1	3	0.172407

[281]:
```
# Display leverage plots
fig = plt.figure(figsize=(12, 6))
sm.graphics.plot_partregress_grid(PayReg2, fig=fig)
plt.show()
```



In light of the relatively low p-value for the Ramsey test with one fitted term, I included the square of the logarithm of employment. It's p-value is .064, and hence it is relatively significant. I also investigated including the square of logMktCap, but the coefficient was very small and not statistically significant.

I included an indicator variable for gender, the logarithm of years as CEO, and the CEO Age. Note that the coefficient of Gender is small and has a high p-value (.33). Hence, this regression implies that, controlling for company size (market cap and number of employees), there is no material difference in pay between male and female CEOs. CEO age is also insignificant. Moreover, and somewhat surprisingly, CEO pay is not significantly related to the number of years the individual has been CEO. The same is true below when indicator variables for industry are included in the model. The coefficients of logMktCap and the two employment variables are virtually the same with or without the gender and CEO variables. Hence, in the interest of space, I do not report the regression without those variables.

[282]:
```
PayReg3=tprstats.model("cs",␣
↪"log(TDC1)~log(Employees)+log(MktCap)+log(YRS_CEO)+ GENDER", USFirms)
PayReg3.summary()
```

[282]:

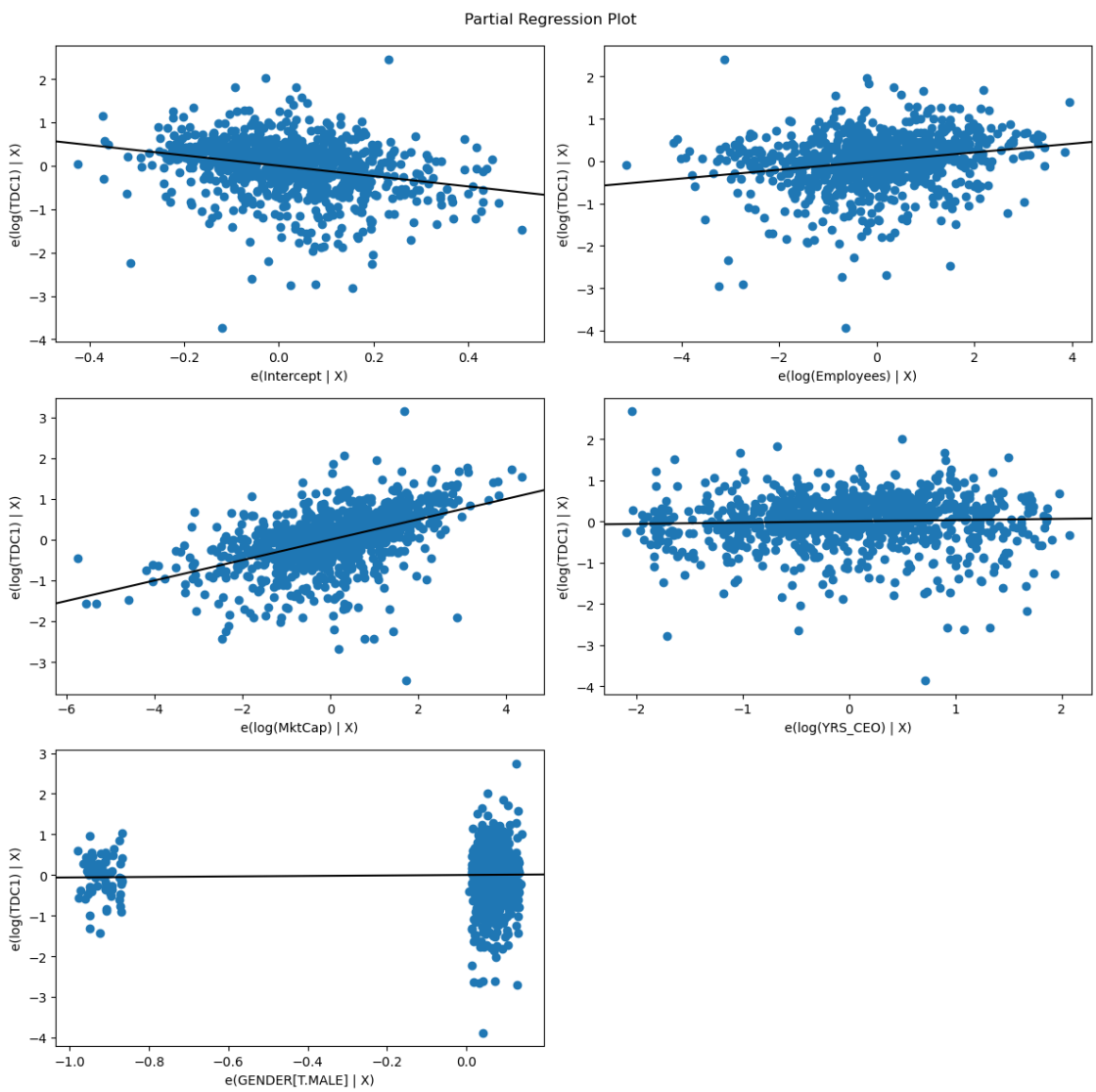
Dep. Variable:	log(TDC1)	**R-squared:**	0.452
Model:	OLS	**Adj. R-squared:**	0.450
No. Observations:	992	**F-statistic:**	190.2
Covariance Type:	HC1	**Prob (F-statistic):**	7.33e-121

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-1.1921	0.139	-8.581	0.000	-1.465	-0.919
log(Employees)	0.1030	0.016	6.294	0.000	0.071	0.135
log(MktCap)	0.2503	0.015	17.026	0.000	0.221	0.279
log(YRS_CEO)	0.0307	0.025	1.225	0.221	-0.019	0.080
GENDER[T.MALE]	0.0607	0.062	0.975	0.330	-0.061	0.183

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Let's view leverage plots of the model.

[283]:
```
# Display leverage plots
fig = plt.figure(figsize=(12,12))
sm.graphics.plot_partregress_grid(PayReg3, fig=fig)
plt.show()
```



Visually, thre is greater number of observations below relative to above the line in the upper right plot. Hence, I decided to try including the square of the logarithm of employees.

[284]:
```
PayReg4=tprstats.model("cs",␣
↪"log(TDC1)~log(Employees)+I(log(Employees)**2)+log(MktCap)+log(YRS_CEO)+AGE",␣
↪USFirms)
PayReg4.summary()
```

[284]:

Dep. Variable:	log(TDC1)	**R-squared:**	0.454
Model:	OLS	**Adj. R-squared:**	0.451
No. Observations:	992	**F-statistic:**	161.4
Covariance Type:	HC1	**Prob (F-statistic):**	2.28e-125

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-2.0230	0.496	-4.082	0.000	-2.995	-1.050
log(Employees)	0.2950	0.109	2.708	0.007	0.081	0.509
I(log(Employees) ** 2)	-0.0112	0.006	-1.879	0.061	-0.023	0.000
log(MktCap)	0.2572	0.015	17.325	0.000	0.228	0.286
log(YRS_CEO)	0.0303	0.028	1.072	0.284	-0.025	0.086
AGE	0.0006	0.004	0.165	0.869	-0.007	0.008

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
[2] The condition number is large, 2.42e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

The small standardized coefficients for log(YRs_CEO) and AGE reinforce the inference that those variables are not very important predictors. In what follows, I retain those variables, but their presence or absence makes litle difference.

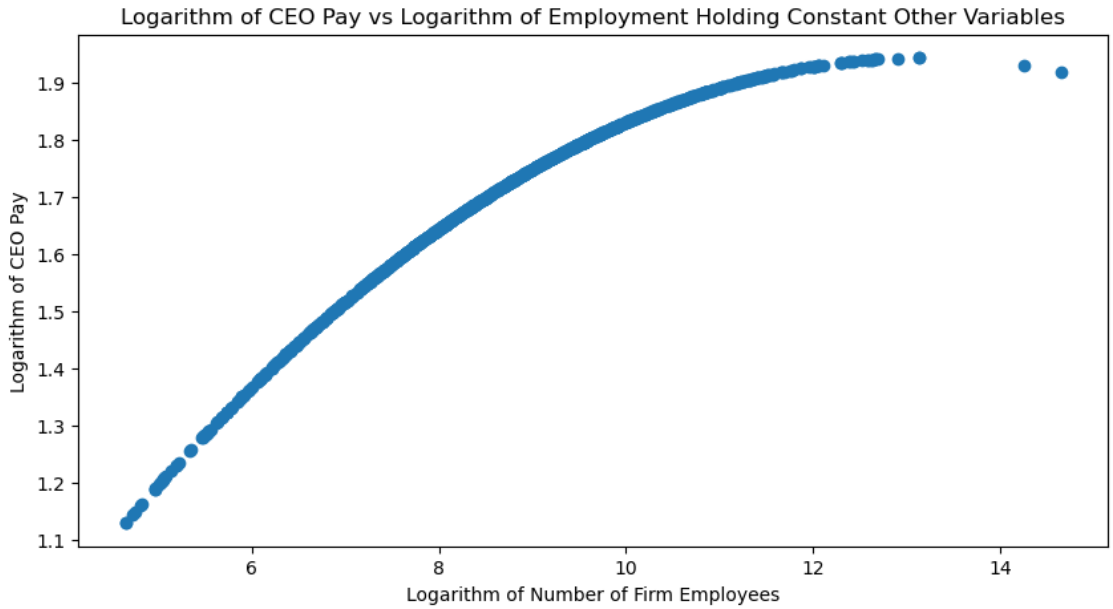
[285]:
```
PayReg4.standardized_coefficients()
```

[285]:
```
log(Employees)            0.604164
I(log(Employees) ** 2)   -0.405707
log(MktCap)               0.537881
log(YRS_CEO)              0.032827
AGE                       0.005189
dtype: float64
```

The graph below displays the terms in the logarithm of employment. We see that the curve increases at a decreasing rate and peaks where the log or employment is roughly 13.

[286]:
```
plt.figure(figsize=(10, 5))
Employees=USFirms["Employees"]
log_Emp=numpy.log(Employees)
log_Emp_sq=log_Emp**2
Emp_pred=.2950*log_Emp-.0112*log_Emp_sq
plt.scatter(log_Emp,Emp_pred)
plt.title('Logarithm of CEO Pay vs Logarithm of Employment Holding Constant␣
↪Other Variables')
plt.xlabel('Logarithm of Number of Firm Employees')
plt.ylabel('Logarithm of CEO Pay')
```

```
plt.show()
```



1.11 Variation in Executive Compensation Across Industries

It is of much interest to investigate the extent to which there is variation in pay across industries. Each US firm is classified by the US government with a four-digit SIC code. The first two digits of the code are the industry in which the firm operates. To study whether there is significant difference in CEO pay across industries, we include a separate indicator variable for each industry. Hence, I used the factor() command to include an indicator variable for each industry. As the reference category, I chose SIC code 36 (Electronic & Other Electric Equipment). There are 55 firms in our sample with this SIC code, including Apple. The coefficients of the market capitalization, employment, and gender variables for the model below are very close in magnitude to the regression above.

Let's study the industry indicator variables. To facilitate interpretation, I selected the industries for which there are at least 10 firms in our sample and for which p-values of the coefficients are less than .051. I also calculated the percent pay difference (PctPayDiff) relative to the reference industry (SIC code 36). The results are in the table below. As you can see, there are large differences in CEO pay across industries holding constant market capitalization and number of employees.[1] In the highly paid group, the three most prominent are General Building Contractors, Communications, and Security & Commodity Brokers, with premia relative to the reference industry of 81%, 65%, and 58% respectively. On the opposite side, Electric & Gas & Sanitary Services, Depository Institutions, and Trucking & Warehousing pay are lower by 19%, 23%, and 38% respectively than the reference industry.

[1] We have controlled for market capitalization and number of employees in the regression.

[287]:
```
# Include indicator variables for industry
# Let's make industry 36 the reference by switching its number to 1
USFirms['SIC2'] = USFirms['SIC2'].replace(36, 1)
PayReg6=tprstats.model("cs",␣
 ↪"log(TDC1)~log(Employees)+I(log(Employees)**2)+log(MktCap)+log(YRS_CEO)+AGE+C(SIC2)",␣
 ↪USFirms)
# Now restore SIC value for industry 36
USFirms['SIC2'] = USFirms['SIC2'].replace(1, 36)
```

[288]:
```
print(PayReg6.summary2())
```

```
                    Results: Ordinary least squares
=====================================================================
Model:                 OLS               Adj. R-squared:      0.486
Dependent Variable:    log(TDC1)         AIC:                 1813.3383
Date:                  2026-03-10 18:27  BIC:                 2126.9205
No. Observations:      992               Log-Likelihood:      -842.67
Df Model:              63                F-statistic:         20.19
Df Residuals:          928               Prob (F-statistic):  2.78e-133
R-squared:             0.518             Scale:               0.34224
---------------------------------------------------------------------
                        Coef.  Std.Err.    t    P>|t|   [0.025  0.975]
---------------------------------------------------------------------
Intercept              -2.2161   0.5455 -4.0625 0.0001 -3.2866 -1.1455
log(Employees)          0.3292   0.1176  2.7987 0.0052  0.0984  0.5601
I(log(Employees) ** 2) -0.0119   0.0064 -1.8615 0.0630 -0.0245  0.0006
log(MktCap)             0.2528   0.0190 13.3162 0.0000  0.2155  0.2900
log(YRS_CEO)            0.0177   0.0281  0.6317 0.5277 -0.0374  0.0728
AGE                     0.0008   0.0040  0.2090 0.8345 -0.0069  0.0086
C(SIC2)[T.10]          -0.0806   0.1119 -0.7203 0.4715 -0.3002  0.1390
C(SIC2)[T.12]           0.3833   0.2339  1.6389 0.1016 -0.0757  0.8424
C(SIC2)[T.13]           0.2718   0.1247  2.1795 0.0295  0.0271  0.5166
C(SIC2)[T.14]           0.3292   0.1637  2.0103 0.0447  0.0078  0.6505
C(SIC2)[T.15]           0.5630   0.1544  3.6459 0.0003  0.2599  0.8660
C(SIC2)[T.16]          -0.2163   0.2529 -0.8554 0.3926 -0.7127  0.2800
C(SIC2)[T.17]          -0.3039   0.1151 -2.6405 0.0084 -0.5297 -0.0780
C(SIC2)[T.20]          -0.0457   0.1228 -0.3720 0.7100 -0.2867  0.1953
C(SIC2)[T.22]          -0.2432   0.2387 -1.0187 0.3086 -0.7117  0.2253
C(SIC2)[T.23]           0.0174   0.2121  0.0820 0.9346 -0.3988  0.4336
C(SIC2)[T.24]          -0.0042   0.1868 -0.0224 0.9821 -0.3708  0.3624
C(SIC2)[T.25]           0.1410   0.2269  0.6216 0.5344 -0.3042  0.5862
C(SIC2)[T.26]           0.1889   0.1742  1.0839 0.2787 -0.1531  0.5308
C(SIC2)[T.27]           0.3147   0.2578  1.2208 0.2225 -0.1912  0.8205
C(SIC2)[T.28]           0.0512   0.1172  0.4366 0.6625 -0.1788  0.2812
C(SIC2)[T.29]           0.0844   0.1973  0.4277 0.6690 -0.3028  0.4716
C(SIC2)[T.30]          -0.0804   0.1450 -0.5543 0.5795 -0.3649  0.2042
C(SIC2)[T.31]           0.3517   0.0954  3.6870 0.0002  0.1645  0.5390
C(SIC2)[T.32]           0.3009   0.1614  1.8638 0.0627 -0.0159  0.6177
```

C(SIC2)[T.33]	0.2740	0.1404	1.9516	0.0513	-0.0015	0.5494
C(SIC2)[T.34]	-0.1783	0.1823	-0.9778	0.3284	-0.5361	0.1795
C(SIC2)[T.35]	0.0473	0.1055	0.4485	0.6539	-0.1597	0.2544
C(SIC2)[T.37]	-0.0249	0.1458	-0.1707	0.8645	-0.3110	0.2612
C(SIC2)[T.38]	-0.1284	0.1270	-1.0112	0.3122	-0.3777	0.1208
C(SIC2)[T.39]	0.3030	0.1738	1.7435	0.0816	-0.0381	0.6441
C(SIC2)[T.40]	-0.1669	0.1706	-0.9785	0.3281	-0.5017	0.1678
C(SIC2)[T.42]	-0.5007	0.2041	-2.4538	0.0143	-0.9012	-0.1002
C(SIC2)[T.44]	-0.0397	0.0988	-0.4017	0.6880	-0.2336	0.1542
C(SIC2)[T.45]	-0.3760	0.2207	-1.7034	0.0888	-0.8091	0.0572
C(SIC2)[T.47]	-0.2803	0.1488	-1.8831	0.0600	-0.5724	0.0118
C(SIC2)[T.48]	0.4971	0.1663	2.9888	0.0029	0.1707	0.8235
C(SIC2)[T.49]	-0.2217	0.1071	-2.0705	0.0387	-0.4318	-0.0116
C(SIC2)[T.50]	-0.3124	0.2106	-1.4831	0.1384	-0.7257	0.1010
C(SIC2)[T.51]	0.0493	0.1286	0.3829	0.7019	-0.2032	0.3017
C(SIC2)[T.52]	-0.3078	0.1377	-2.2349	0.0257	-0.5781	-0.0375
C(SIC2)[T.53]	0.0261	0.2178	0.1197	0.9048	-0.4013	0.4534
C(SIC2)[T.55]	-0.0476	0.1591	-0.2994	0.7647	-0.3599	0.2646
C(SIC2)[T.56]	-0.0905	0.2683	-0.3372	0.7360	-0.6171	0.4361
C(SIC2)[T.57]	0.2416	0.2606	0.9273	0.3540	-0.2698	0.7530
C(SIC2)[T.58]	-0.2744	0.1567	-1.7511	0.0803	-0.5819	0.0331
C(SIC2)[T.59]	-0.1951	0.2633	-0.7409	0.4590	-0.7119	0.3217
C(SIC2)[T.60]	-0.2791	0.1212	-2.3040	0.0214	-0.5169	-0.0414
C(SIC2)[T.61]	0.3774	0.1366	2.7629	0.0058	0.1093	0.6454
C(SIC2)[T.62]	0.4507	0.1381	3.2635	0.0011	0.1797	0.7218
C(SIC2)[T.63]	0.0079	0.1360	0.0580	0.9537	-0.2590	0.2748
C(SIC2)[T.64]	-0.0532	0.1234	-0.4312	0.6665	-0.2953	0.1889
C(SIC2)[T.65]	0.1438	0.2565	0.5607	0.5752	-0.3596	0.6472
C(SIC2)[T.67]	0.1400	0.1467	0.9547	0.3400	-0.1478	0.4279
C(SIC2)[T.70]	-0.0988	0.2393	-0.4129	0.6798	-0.5684	0.3708
C(SIC2)[T.72]	0.3086	0.3023	1.0206	0.3077	-0.2848	0.9019
C(SIC2)[T.73]	0.0765	0.1187	0.6446	0.5194	-0.1564	0.3095
C(SIC2)[T.75]	0.2384	0.4985	0.4783	0.6325	-0.7399	1.2168
C(SIC2)[T.78]	0.3703	0.2541	1.4571	0.1454	-0.1284	0.8691
C(SIC2)[T.79]	-0.1881	0.2034	-0.9245	0.3555	-0.5873	0.2112
C(SIC2)[T.80]	-0.0303	0.2046	-0.1483	0.8821	-0.4318	0.3711
C(SIC2)[T.82]	-0.0348	0.1899	-0.1830	0.8549	-0.4075	0.3380
C(SIC2)[T.87]	-0.0458	0.1768	-0.2589	0.7957	-0.3927	0.3012
C(SIC2)[T.99]	-0.3979	0.4180	-0.9520	0.3414	-1.2183	0.4224

Omnibus:	232.459	Durbin-Watson:	2.017
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1068.329
Skew:	-1.013	Prob(JB):	0.000
Kurtosis:	7.663	Condition No.:	3661

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
[2] The condition number is large, 3.66e+03. This might indicate

that there are strong multicollinearity or other numerical problems.

[289]:
```
# Calculate Differences Across Industry
# Highest
100*(numpy.exp(.5630)-1)
```

[289]: 75.5932

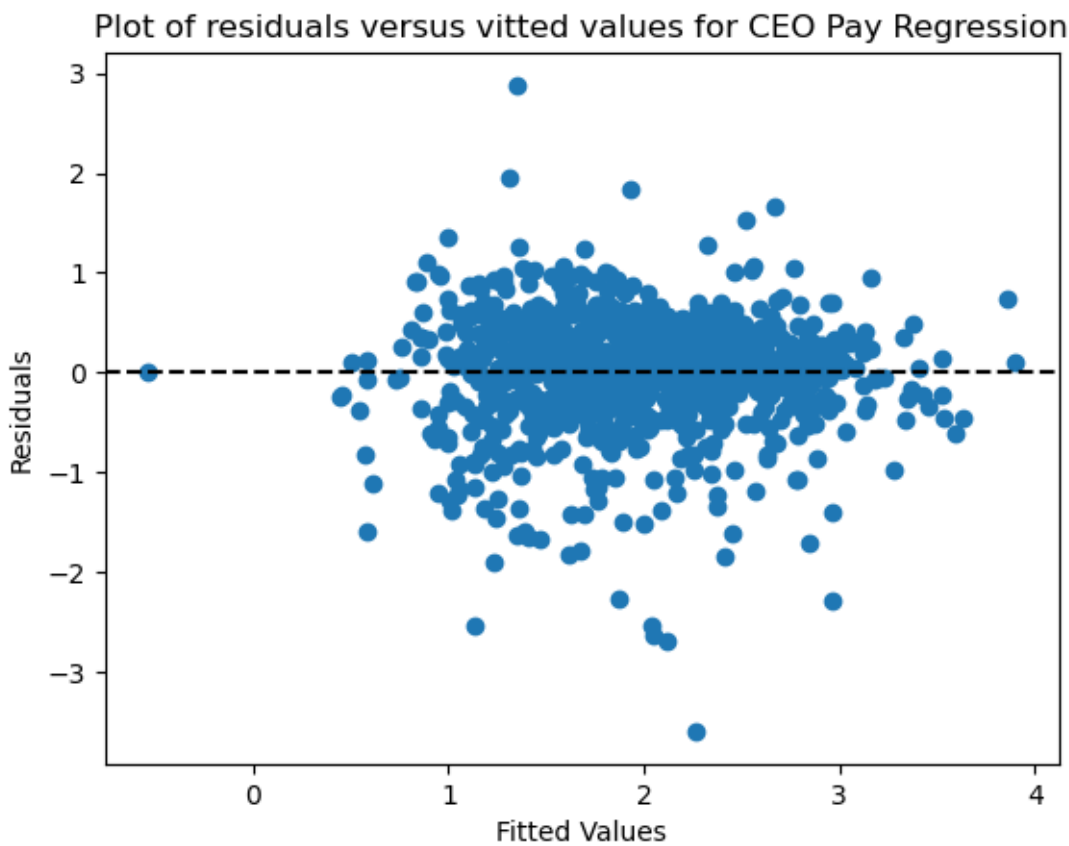
[290]:
```
# Calculate Differences Across Industry
# Lowest
100*(numpy.exp(-.5007)-1)
```

[290]: -39.3894

A plot of the residuals versus the fitted values yields no visible evidence of a nonlinear relationship.

[291]:
```
# Extract residuals and fitted values and do plot
resid=PayReg6.resid
yhat=PayReg6.fittedvalues
plt.scatter(yhat,resid)
plt.axhline(y=0, color='black', linestyle='--')
plt.title('Plot of residuals versus vitted values for CEO Pay Regression')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.show
```

[291]: <function matplotlib.pyplot.show(close=None, block=None)>



[292]:
```
# Find the smallest and largest residuals. Recall that Python numbers␣
↪observations starting at zero.
# Hence, 1 is added below since the numbering of observations in the data set␣
↪begins with 1.
# The smallest (i.e., most negative) residual is observation 235 and the␣
↪largest is observation 725
# These are respectively Copart and Parker-Hannafin.
# Greatest underprediction is for the CEO of Copart and greatest overprediction␣
↪is for CEO of Parker-Hannafin

idxmin = resid.idxmin()
idxmax = resid.idxmax()
min_val = resid.loc[idxmin]
[idxmin+1,idxmax+1]
```

[292]: [235, 725]

The Ramsey test yields high p-values, suggesting that this is a much improved functional form.

[293]:
```
PayReg6.ramsey_test()
```

[293]:

```
   power    pvalue
0      2  0.198944
1      3  0.406056
```

[294]:

```
FixEfDat = pandas.read_excel("data/SIC2_FE_and_Description.xlsx")
```

[295]:

```
KeepFixe = pandas.read_excel("data/KeepFixe.xlsx")
```

[296]:

```
KeepFixe = FixEfDat[(FixEfDat['pval'] < .05) & (FixEfDat['Count'] > 9)]
KeepFixe = KeepFixe.drop(columns=['stderr', 'tstat'])
print(KeepFixe)
```

```
    SIC2      Coef      pval  Count  PctPayDiff                      Activity
1     13  0.343261  0.000654     26   40.953660          Oil & Gas Extraction
3     15  0.648454  0.000002     15   91.258169  General Building Contractors
18    33  0.353585  0.003931     14   42.416404      Primary Metal Industries
26    42 -0.417411  0.034726     10  -34.124988        Trucking & Warehousing
30    48  0.579477  0.000104     18   78.510458                Communications
41    60 -0.199424  0.036848     65  -18.079752       Depository Institutions
42    61  0.458623  0.000103     10   58.189422    Nondepository Institutions
43    62  0.537206  0.000005     20   71.121903  Security & Commodity Brokers
```

1.12 Summary

The above analysis provides you with an overview of firms in the US economy. The analysis also demonstrates the use of our software for summarizing data, fitting distributions to data, and conducting regression analysis. In addition, the analysis demonstrates the value of expressing variables in log form when the distributions of the variables are highly skewed. The regression in log form with industry indicator variables (Appendix C) has an adjusted R^2 value of .49. Thus, the logarithms of employment, market capitalization, and industry indicator variables predict roughly half of the variation in CEO compensation for the 992 firms in our data.

1.13 Appendix A: Table of SIC 2-digit Industries

[297]:
```
pandas.set_option('display.max_rows', None)
SIC2_and_Descriptions_Numbers = pandas.read_excel("data/
↪SIC2_and_Descriptions_Numbers.xlsx")
print(SIC2_and_Descriptions_Numbers)
```

```
    SIC2                                 Activity
0      1            Agricultural Production & Crops
1      2        Agricultural Production & Livestock
2      7                      Agricultural Services
3      8                                  Forestry
4      9               Fishing, Hunting, & Trapping
5     10                              Metal, Mining
6     12                                Coal Mining
7     13                       Oil & Gas Extraction
8     14         Nonmetallic Minerals, Except Fuels
9     15               General Building Contractors
10    16        Heavy Construction, Except Building
11    17                  Special Trade Contractors
12    20                    Food & Kindred Products
13    21                           Tobacco Products
14    22                      Textile Mill Products
15    23           Apparel & Other Textile Products
16    24                      Lumber & Wood Products
17    25                        Furniture & Fixtures
18    26                    Paper & Allied Products
19    27                      Printing & Publishing
20    28                 Chemical & Allied Products
21    29                  Petroleum & Coal Products
22    30   Rubber & Miscellaneous Plastics Products
23    31                 Leather & Leather Products
24    32              Stone, Clay, & Glass Products
25    33                   Primary Metal Industries
26    34                  Fabricated Metal Products
27    35           Industrial Machinery & Equipment
28    36      Electronic & Other Electric Equipment
29    37                    Transportation Equipment
30    38             Instruments & Related Products
31    39     Miscellaneous Manufacturing Industries
32    40                     Railroad Transportation
33    41       Local & Interurban Passenger Transit
34    42                      Trucking & Warehousing
35    43                           US Postal Service
36    44                        Water Transportation
37    45                      Transportation by Air
38    46              Pipelines, Except Natural Gas
39    47                    Transportation Services
```

40	48	Communications
41	49	Electric, Gas, & Sanitary Services
42	50	Wholesale Trade & Durable Goods
43	51	Wholesale Trade & Nondurable Goods
44	52	Building Materials & Gardening Supplies
45	53	General Merchandise Stores
46	54	Food Stores
47	55	Automotive Dealers & Service Stations
48	56	Apparel & Accessory Stores
49	57	Furniture & Homefurnishings Stores
50	58	Eating & Drinking Places
51	59	Miscellaneous Retail
52	60	Depository Institutions
53	61	Nondepository Institutions
54	62	Security & Commodity Brokers
55	63	Insurance Carriers
56	64	Insurance Agents, Brokers, & Service
57	65	Real Estate
58	67	Holding & Other Investment Offices
59	70	Hotels & Other Lodging Places
60	72	Personal Services
61	73	Business Services
62	75	Auto Repair, Services, & Parking
63	76	Miscellaneous Repair Services
64	78	Motion Pictures
65	79	Amusement & Recreation Services
66	80	Health Services
67	81	Legal Services
68	82	Educational Services
69	83	Social Services
70	84	Museums, Botanical, Zoological Gardens
71	86	Membership Organizations
72	87	Engineering & Management Services
73	88	Private Households
74	89	Services, Not Elsewhere Classified
75	91	Executive, Legislative, & General
76	92	Justice, Public Order, & Safety
77	93	Finance, Taxation, & Monetary Policy
78	94	Administration of Human Resources
79	95	Environmental Quality & Housing
80	96	Administration of Economic Programs
81	97	National Security & International Affairs
82	98	Zoological Gardens
83	99	Non-Classifiable Establishments

The following table lists the number of firms in each industry in our sample. Recall that our data set is made up of firms that have more than 99 employees and also have CEO compensation more than $99,000. Some industries do not appear in our sample because they do not have any firms

that meet these two criteria.

1.14 Appendix B: Number of Firms By Industry in Our Sample

[298]:
```
Count_SIC2_and_Activity = pandas.read_excel("data/Count_SIC2_and_Activity.xlsx")
Count_SIC2_and_Activity
```

[298]:

	SIC2	Number	Activity
0	10	3	Metal, Mining
1	12	5	Coal Mining
2	13	26	Oil & Gas Extraction
3	14	6	Nonmetallic Minerals, Except Fuels
4	15	15	General Building Contractors
5	16	6	Heavy Construction, Except Building
6	17	5	Special Trade Contractors
7	20	29	Food & Kindred Products
8	22	4	Textile Mill Products
9	23	5	Apparel & Other Textile Products
10	24	5	Lumber & Wood Products
11	25	7	Furniture & Fixtures
12	26	8	Paper & Allied Products
13	27	3	Printing & Publishing
14	28	68	Chemical & Allied Products
15	29	7	Petroleum & Coal Products
16	30	6	Rubber & Miscellaneous Plastics Products
17	31	3	Leather & Leather Products
18	32	2	Stone, Clay, & Glass Products
19	33	14	Primary Metal Industries
20	34	13	Fabricated Metal Products
21	35	61	Industrial Machinery & Equipment
22	36	55	Electronic & Other Electric Equipment
23	37	35	Transportation Equipment
24	38	61	Instruments & Related Products
25	39	7	Miscellaneous Manufacturing Industries
26	40	3	Railroad Transportation
27	42	10	Trucking & Warehousing
28	44	3	Water Transportation
29	45	6	Transportation by Air
30	47	2	Transportation Services
31	48	18	Communications
32	49	63	Electric, Gas, & Sanitary Services
33	50	23	Wholesale Trade & Durable Goods
34	51	13	Wholesale Trade & Nondurable Goods
35	52	4	Building Materials & Gardening Supplies
36	53	7	General Merchandise Stores
37	55	11	Automotive Dealers & Service Stations
38	56	7	Apparel & Accessory Stores

39	57	3	Furniture & Homefurnishings Stores
40	58	16	Eating & Drinking Places
41	59	8	Miscellaneous Retail
42	60	65	Depository Institutions
43	61	10	Nondepository Institutions
44	62	20	Security & Commodity Brokers
45	63	32	Insurance Carriers
46	64	4	Insurance Agents, Brokers, & Service
47	65	8	Real Estate
48	67	57	Holding & Other Investment Offices
49	70	6	Hotels & Other Lodging Places
50	72	4	Personal Services
51	73	73	Business Services
52	75	4	Auto Repair, Services, & Parking
53	78	3	Motion Pictures
54	79	7	Amusement & Recreation Services
55	80	20	Health Services
56	82	7	Educational Services
57	87	13	Engineering & Management Services
58	99	2	Non-Classifiable Establishments